Predicting and Analyzing NBA Performance
Question 3 — Playoffs vs. Regular Season

Which players raise their game when the stakes are highest?

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1. Introduction

The NBA playoffs are widely regarded as a different kind of basketball. The pace slows, defenses tighten, and opposing coaches spend days preparing to neutralize individual players. Whether elite regular-season performers maintain or even improve their production in this environment has long been a subject of debate among fans, analysts, and front offices alike.

This analysis uses 28 seasons of data (1995–2023) to answer the question systematically. We work with **4,982 player-seasons** in which the same player appeared in both the regular season and the playoffs, allowing a direct apples-to-apples comparison.

2. Dataset & Methodology

Each observation is a player-season pair from the Kaggle dataset *bendikfltaas/nba-history-seasonal-data-1995-2023*, supplemented with advanced metrics (PER, BPM, VORP, WS/48, TS%, USG%) from Basketball Reference and biographical records (position, age, experience, draft number) from the NBA Stats API.

A player-season is labelled **Riser** if playoff PPG exceeded regular-season PPG beyond a threshold, **Decliner** if it fell below, and **Neutral** otherwise. Of the 4,982 qualifying seasons: Risers 656 (13.2%) · Neutral 2,373 (47.6%) · Decliners 1,953 (39.2%).

Decliners outnumber Risers roughly 3-to-1 — the single most important high-level finding. Most players score *less* in the playoffs than in the regular season.

3. The Scoring Delta Distribution

The average Riser scores **+4.05 ppg more** in the playoffs; the average Decliner scores **4.04 ppg less**. The KDE curves below show both distributions are roughly bell-shaped but the Decliner distribution is noticeably wider and shifted further from zero — players who fall off tend to fall harder than those who rise.

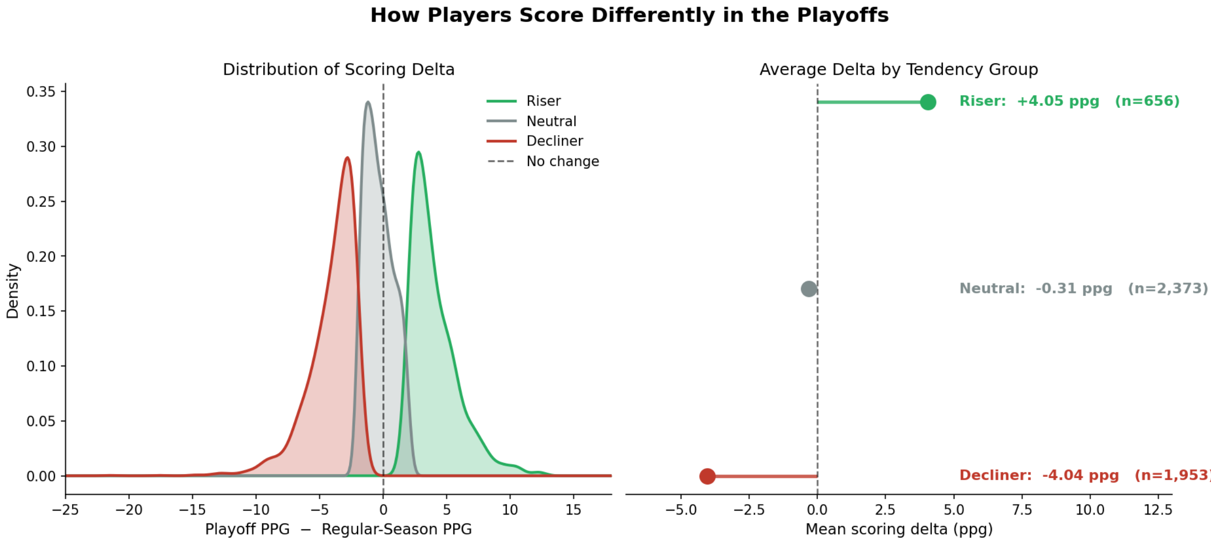

Figure 1. Left: smoothed density curves of playoff scoring delta for each tendency group. Right: mean delta per group shown as a lollipop chart. The dashed vertical line marks zero (no change from regular season).

4. Who Rises? Position and Scoring Role

The stacked charts below show what fraction of each group's player-seasons ends up in each tendency category. Two patterns stand out.

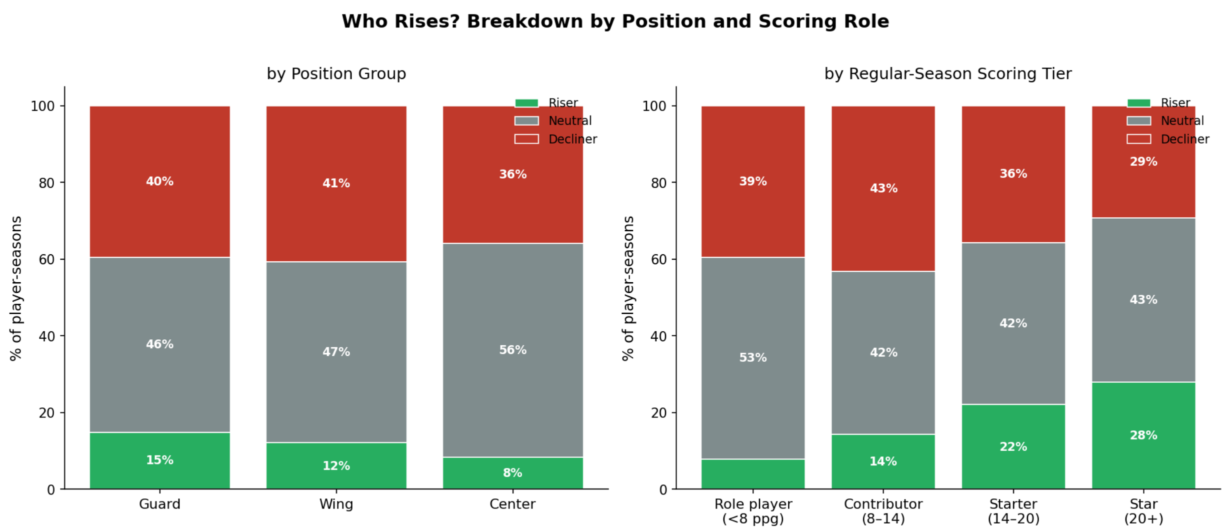

Figure 2. 100% stacked bar charts by position group (left) and regular-season scoring role (right). Each bar sums to 100%. Numbers inside segments show the exact percentage.

By position: Guards have the highest Riser rate (Center: 8% · Guard: 15% · Wing: 12%). Centers are most often Neutral, likely because they operate in more defined roles and are less frequently the primary defensive focus.

By scoring role: The most striking finding is that **star players (20+ ppg) have both the highest Riser rate and the highest Decliner rate** — the widest spread. Role players cluster tightly around Neutral. Stars either step up to the moment or are brought down by the defensive game-plans that specifically target them.

5. Advanced Metric Shifts

Looking beyond points, Risers improve on virtually every advanced metric. Mean PER delta: Risers **+3.61** vs Decliners **-6.65**. The BPM gap is statistically significant (t = 18.88, p = 1.21e-74). Axes are clipped to the central 98% of the distribution to keep outliers from compressing the visible range.

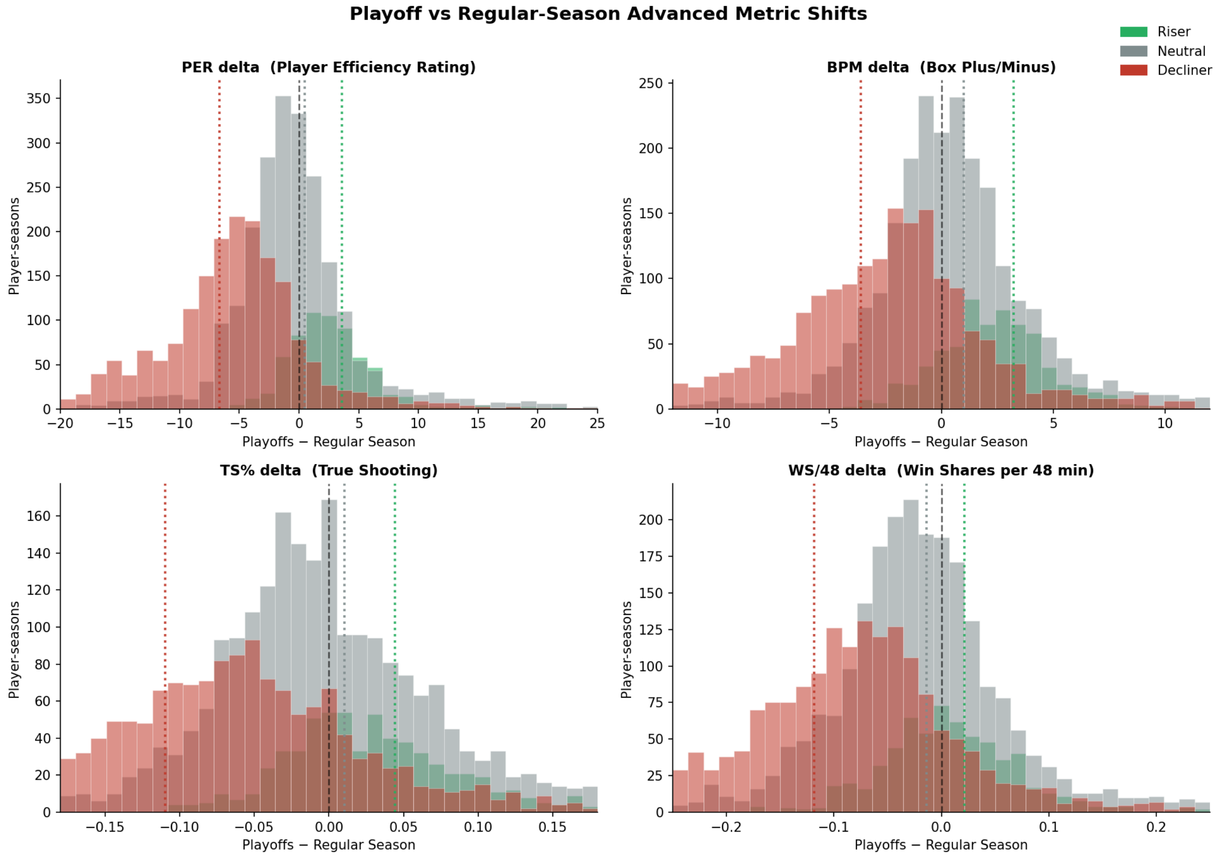

Figure 3. Distribution of advanced metric deltas for each tendency group. Dotted vertical lines mark each group's mean. Axes are clipped to the central distribution — extreme outliers exist but are not shown.

True Shooting % (TS%) delta is positive on average for Risers, meaning they are **more efficient** in the playoffs even when they score more. This is not simply higher volume — it is a genuine step-up in quality.

6. Regular Season vs. Playoff Scoring

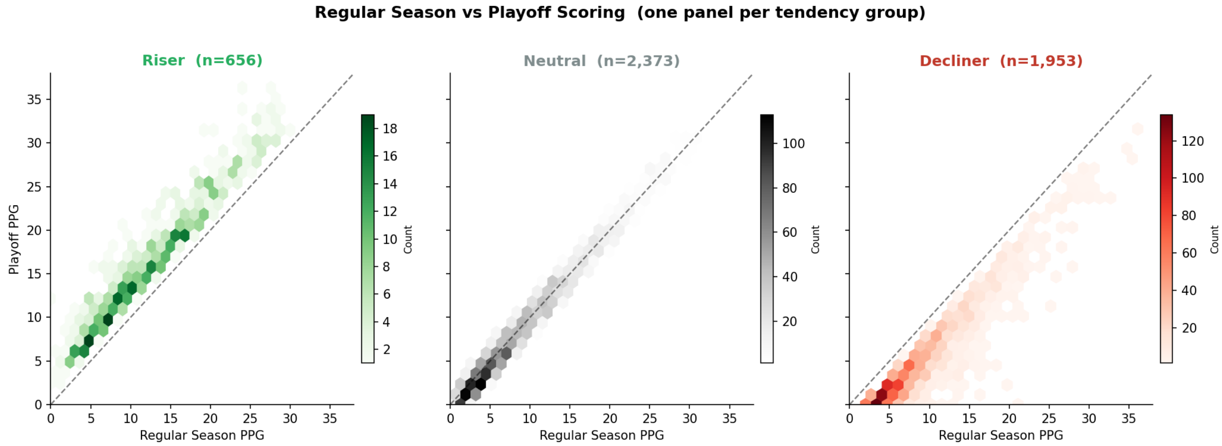

Figure 4. Hexbin density maps — one panel per tendency group. Each hexagonal cell is shaded by the number of player-seasons landing there. The dashed diagonal is the line of equal scoring in both contexts.

The Riser panel shows density concentrated *above* the diagonal. The Decliner panel shows the opposite. The Neutral panel is tightly centred on the diagonal. Points near the origin (role players) tend toward Neutral regardless of group.

7. Serial Risers and Decliners

Some players show the same tendency year after year — suggesting a stable trait, not noise.

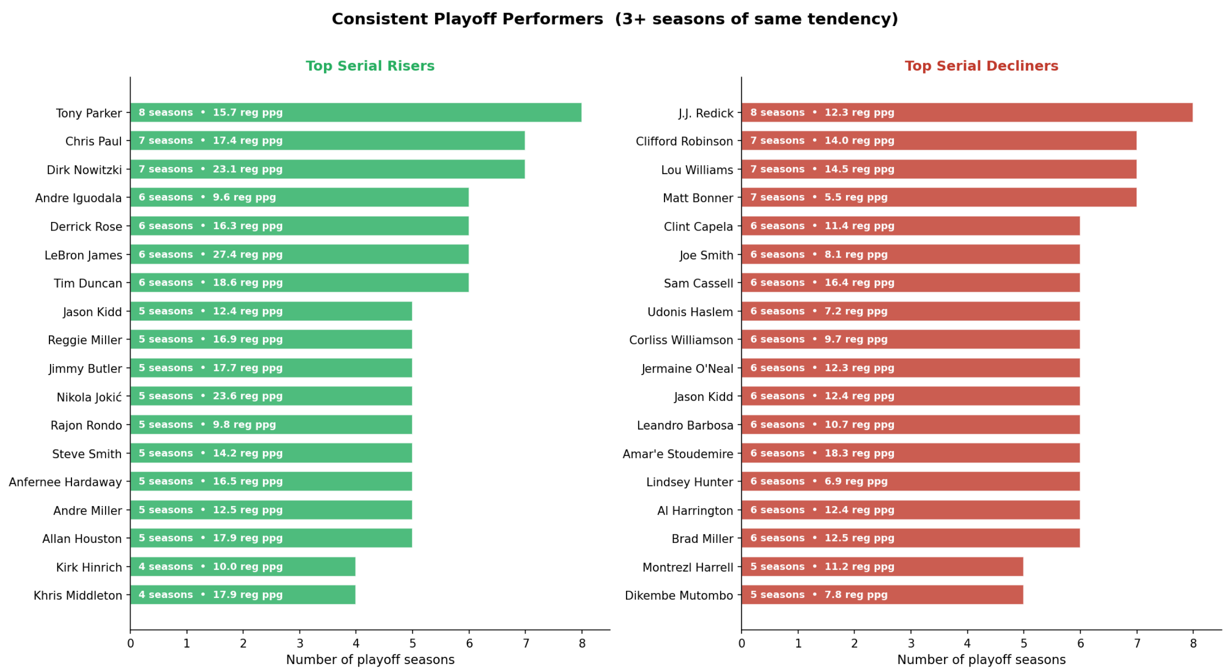

Figure 5. Players with 3+ seasons of the same tendency. Bar labels show season count and career regular-season scoring average.

Top serial Risers: **Tony Parker, Chris Paul, Dirk Nowitzki, Andre Iguodala, Derrick Rose** — widely regarded by analysts as great playoff performers, which validates the methodology.

Top serial Decliners: **J.J. Redick, Clifford Robinson, Lou Williams, Matt Bonner, Clint Capela**. Several were elite regular-season scorers who faced intense playoff defensive attention, consistent with the star-player finding in Section 4.

8. What Predicts a Playoff Riser?

A Random Forest classifier trained on 19 regular-season features distinguished Risers from Decliners with 5-fold CV AUC = **0.70** (σ = 0.03). Moderate but real predictive signal exists in the regular-season data.

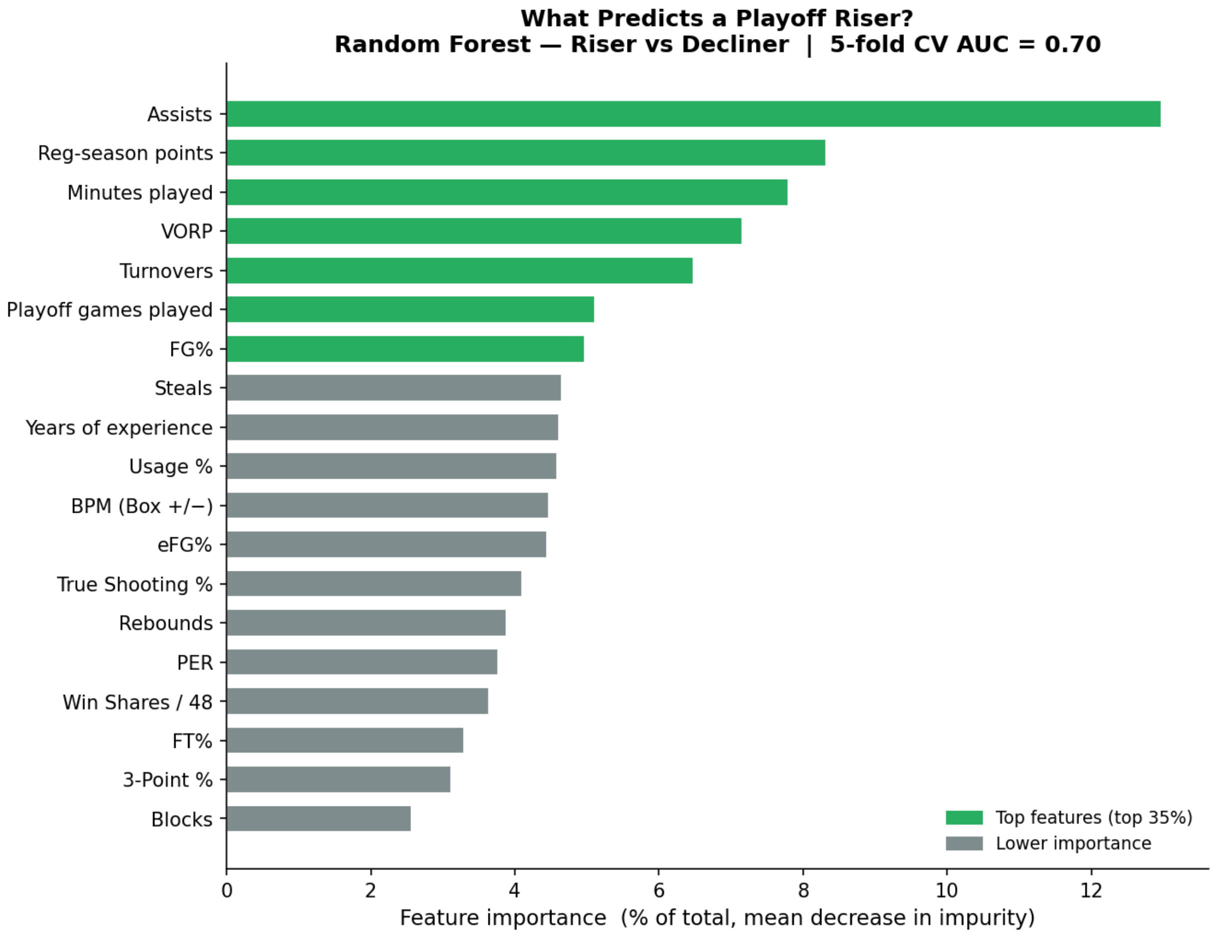

Figure 6. Feature importances as % of total (Random Forest mean decrease in impurity). Green bars = top 35% of features.

Top-5 features: **Assists** (13.0%), **Reg-season points** (8.3%), **Minutes played** (7.8%), **VORP** (7.2%), **Turnovers** (6.5%). Assists and raw volume stats rank highly because they capture overall player involvement — players with a larger footprint in the regular season tend to maintain or grow it in the playoffs. Efficiency metrics (VORP, BPM) beat shooting percentages, suggesting overall positive impact matters more than shooting mechanics.

9. Role Players vs. Stars: Who Takes the Bigger Hit?

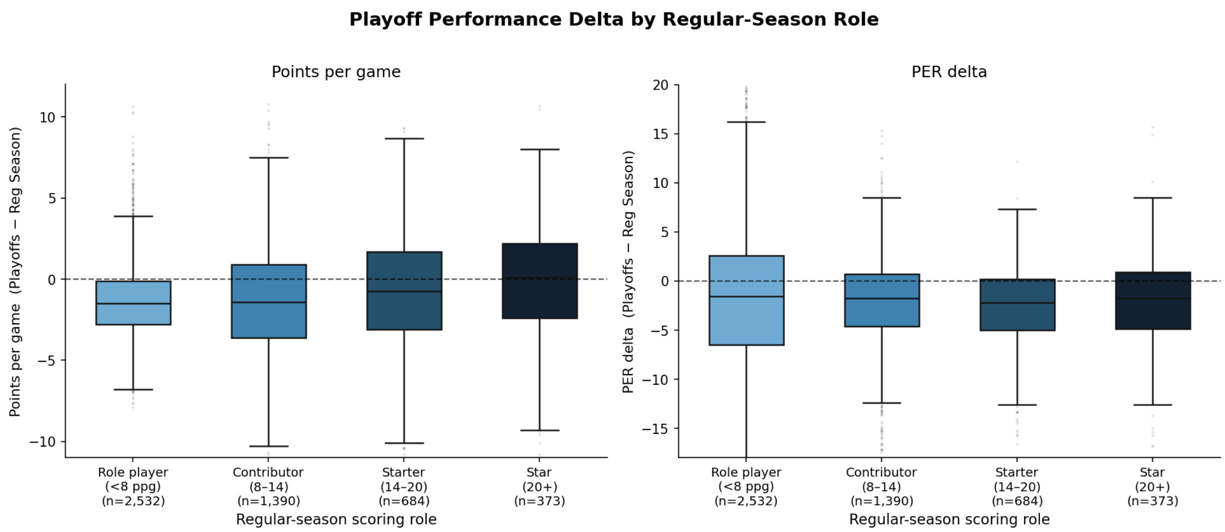

Figure 7. Box plots of scoring delta (left) and PER delta (right) by regular-season scoring tier. Y-axes are clipped to the central distribution range. Dots show outliers within that range.

Stars (20+ ppg) have the widest interquartile range and the highest median — they are the only tier with a positive median scoring delta. But their distribution also extends furthest downward. Role players (<8 ppg) are tightly clustered around zero in both metrics.

10. Does Age or Experience Matter?

We tested whether a player's age or career experience at the time of a playoff run predicts their tendency. Age was approximated as the season year minus birth year.

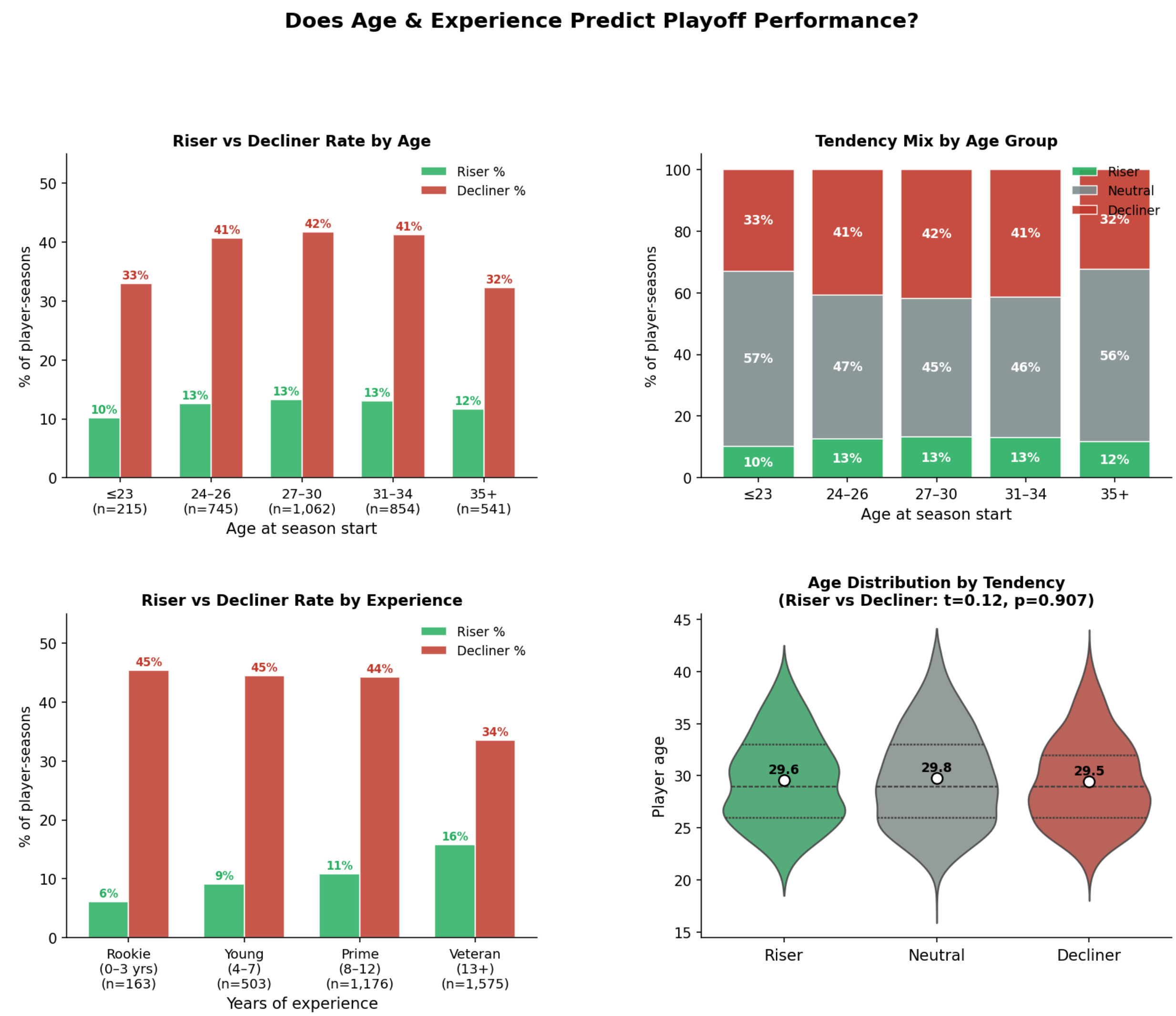

Figure 8. Top row: Riser vs Decliner rate by age group (left) and full tendency mix stacked bars (right). Bottom row: Riser vs Decliner rate by experience tier (left) and violin distributions of player age per tendency group (right, white dots = mean).

Age effect: The mean age of Risers (29.6) versus Decliners (29.5) is similar (t = 0.12, p = 0.907). The difference is not statistically significant, meaning **age alone does not determine playoff tendency**. Young players are not systematically better or worse — the effect is largely flat across age groups.

Experience effect: Veterans (13+ years) show a slightly higher Riser rate on the scoring delta chart, suggesting that playoff experience does provide a small edge. However the effect is modest. The more powerful predictors remain player role and overall efficiency (as shown in Section 8), not age or seniority.

11. Conclusions

1. **Most players score less in the playoffs.** Decliners outnumber Risers 1953:656 (39% vs 13%). Playoff defenses are real.

2. **Stars are the most volatile.** High scorers have both the highest Riser and highest Decliner rates — they polarize, while role players cluster around neutral.

3. **Guards rise most by position.** Centers are the most neutral, likely reflecting their more constrained playoff rotation roles.

4. **True Risers improve on every advanced metric.** PER, BPM, TS%, and WS/48 all shift positively for Risers — they score more *and* more efficiently.

5. **Regular-season data predicts tendency with AUC ≈ 0.70.** Assists, minutes, and efficiency metrics (VORP) are stronger signals than raw scoring volume.

6. **Age alone does not predict playoff tendency.** The Riser vs Decliner age difference is not statistically significant. Player role and efficiency are far more predictive.

Data: Kaggle (bendikfltaas/nba-history-seasonal-data-1995-2023, drgilermo/nba-players-stats), Basketball Reference, NBA Stats API · Coverage: 1995–2023.